

Tiel-Coder on the Ollama box

Benchmark round v4 · 192.168.100.67 · Ollama 0.33.3 · 35.56 GB usable VRAM ·
generated from results/*.tsv

✓ S1 Tiel battery

✓ S2 field on 0.33.3

... S3 sessions

... S5 CyberTiel

Stages still running append to the same files; regenerate with ./make-report.py .

Use **tiel-coder:35b-q5-ctx256k-agentic** , not the tag as shipped. Identical weights; the shipped **-ctx262k** tag carries **presence_penalty 1.5** , which costs **52%** of generation speed and buys nothing measurable.

Tiel holds its full **262,144-token window at 34.13 GB, 100% on GPU**, retrieves a needle at **254,181 tokens** (the deepest in v1–v4), scores **25/25** on the vision checks at that full window, and is the **only model tested that refuses an over-long prompt** instead of silently answering from half of it.

Findings that change what you should do

1 · The runtime now caches prompt prefixes — v3's ranking rule is void

Ollama 0.33.3 reports **cache_read_input_tokens** ; 0.32.15 did not, and every v3 transcript shows zero. A repeated 30k-token prompt prefills **4 tokens instead of 30,042**, and an agent turn (same prefix, new tail) prefills only the tail. v3 ranked models on prefill because "the loop re-reads its context every turn" — that is no longer true here.

2 · The silent half-window bug is still live — but not on Tiel

Past its window, **ornith** , **north-mini** (the v3 default) and the **qwen3.6** control all silently keep **num_ctx/2 + 2** tokens and answer anyway. Tiel returns **HTTP 400**. Claude Code cannot send **num_ctx** , so a model in the halving class can answer from half your context with nothing in the transcript saying so.

3 · **presence_penalty 1.5** is a pure tax

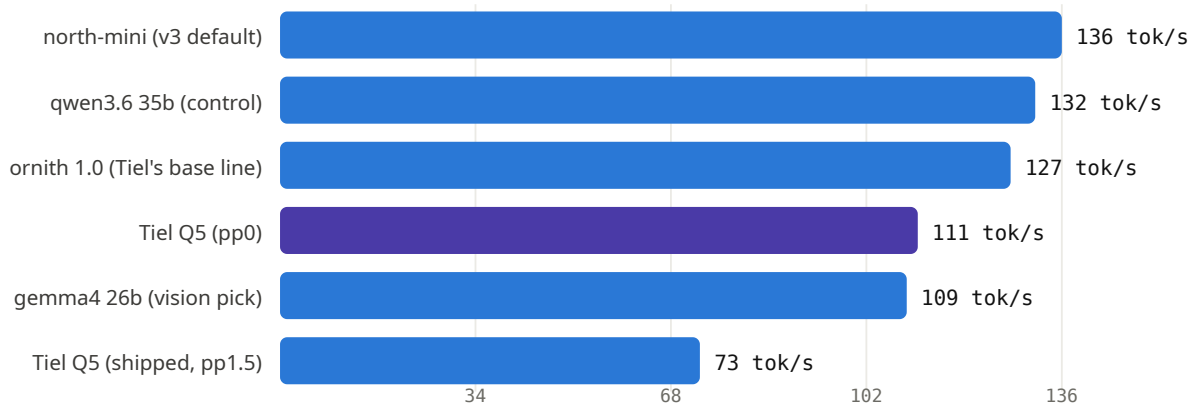
It was added by whoever created the **-ctx262k** tag, not by the model publisher, whose card recommends no penalty at all. It costs 41–52% of generation, leaves prefill unchanged, and makes no measurable difference to tool reliability (T5 re-runs: 13/16 with it, 15/16 without).

4 · A clean gate battery is not a reliability figure

Tiel passed T1–T7 three times over. Re-running the nested-schema gate alone sixteen times gives **13/16**. The battery samples once at the tag's shipped temperature 0.6, so any single-shot gate result in v1–v4 is one sample.

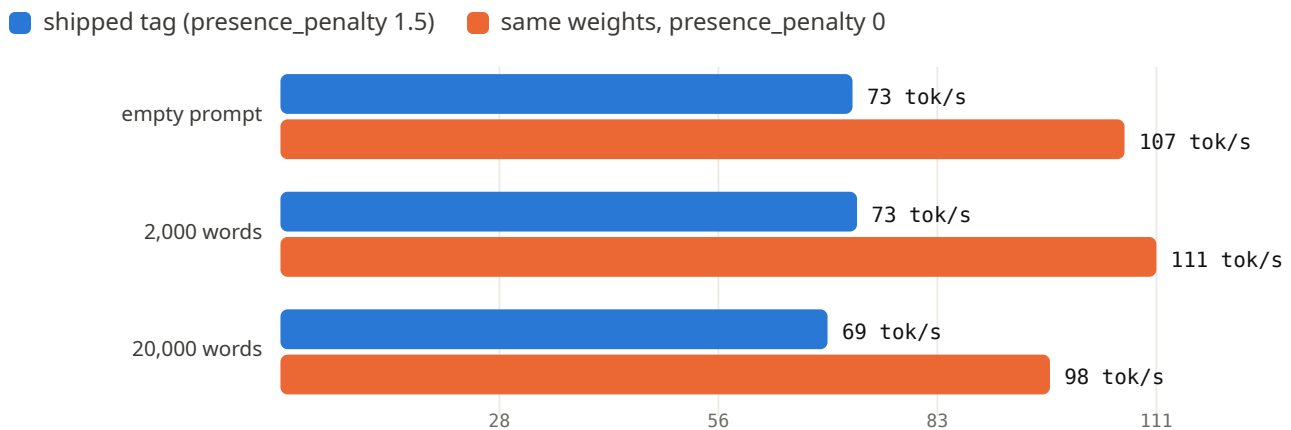
Speed

Generation throughput — 2,000-word prompt



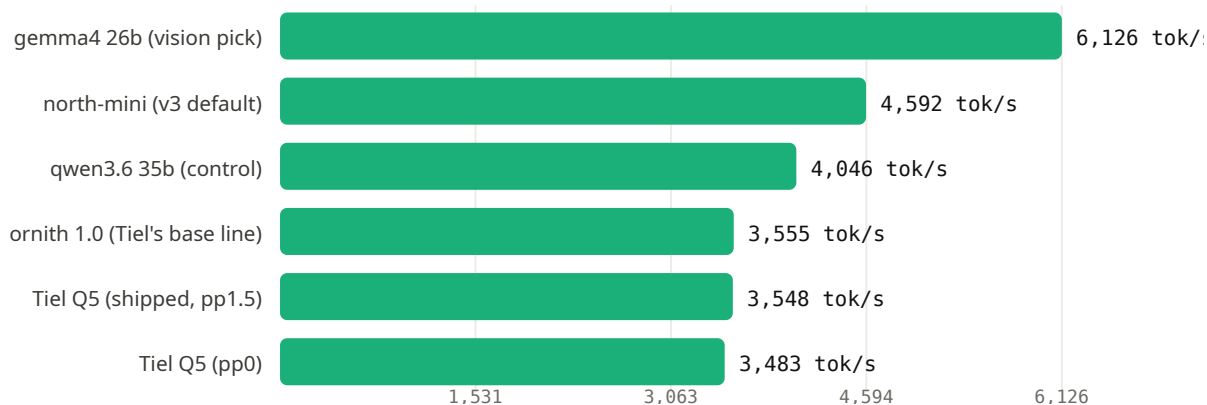
Ollama's own counters, temperature 0, seed 42, thinking off, 256-token budget. Cold prompts. Higher is better.

What presence_penalty 1.5 costs — generation tok/s



Identical weights; only the sampler setting differs. Prefill is unchanged (within 2%), because the penalty is paid per generated token.

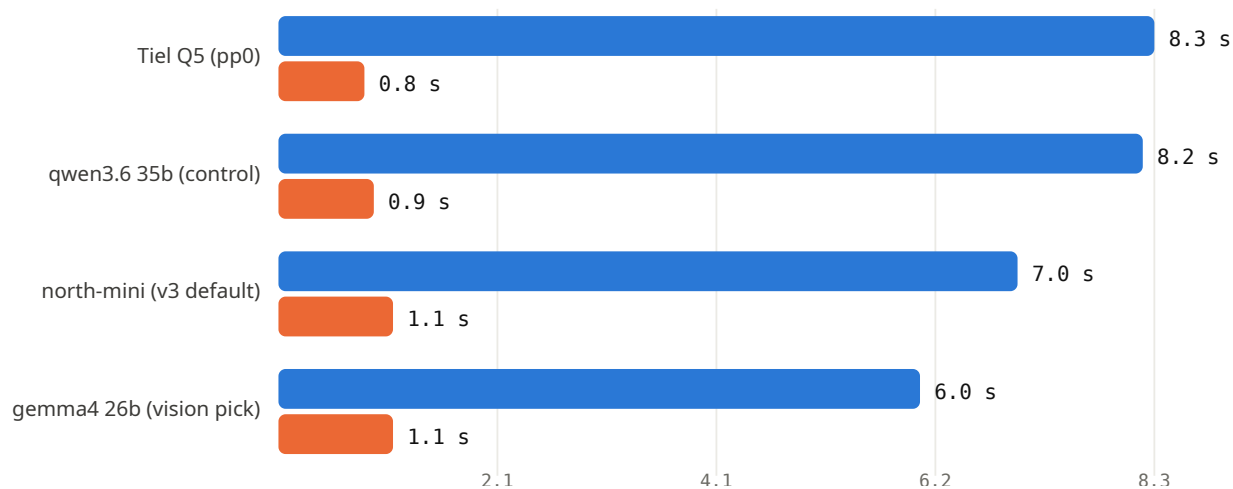
Cold prefill — ~35,000-token prompt



What the FIRST turn of a session costs. Later turns hit the prefix cache instead — see below.

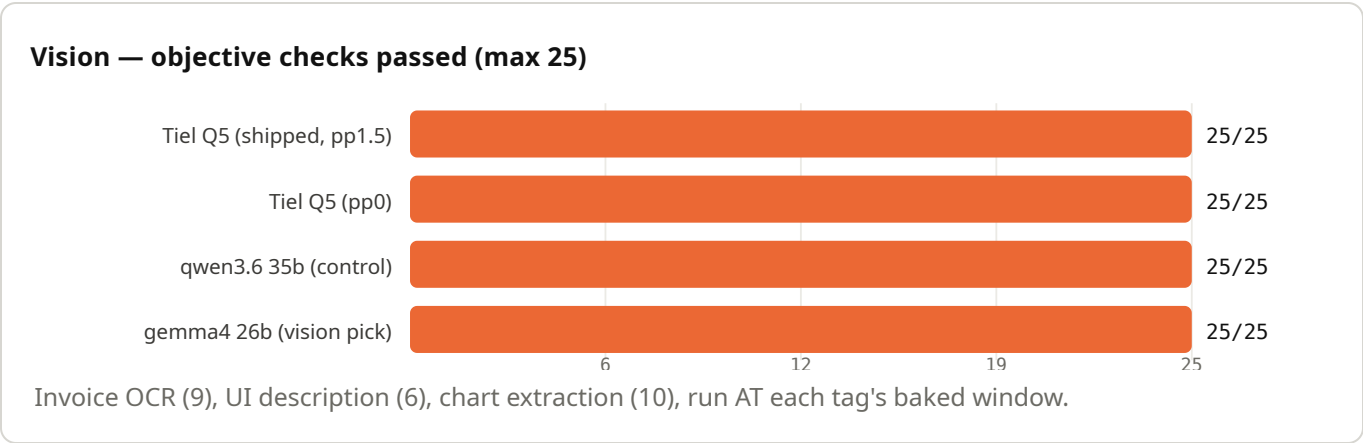
Prefix caching on Ollama 0.33.3 — seconds to first token

■ cold: a new 30k prompt ■ agent turn: same prefix + new tail



0.32.15 had no prefix cache (every v3 transcript reports cache_read 0). An agent turn appends to a stable prefix, so it now prefills only the tail.

Capability



BEHAVIOUR PAST THE CONTEXT WINDOW

MODEL	WHAT HAPPENS WHEN THE PROMPT EXCEEDS NUM_CTX	TOKENS KEPT
ornith 1.0 (Tiel's base line)	✖ silently halves	131074
north-mini (v3 default)	✖ silently halves	131075
Tiel Q5 (shipped, pp1.5)	✔ refuses (HTTP 400)	
qwen3.6 35b (control)	✖ silently halves	131074
gemma4 26b (vision pick)	✖ silently halves	131075

End-to-end Claude Code sessions

Two fixtures against a real `claude -p` session, scored from the repository rather than the model's summary: a one-function bug, and a three-module `ledger` package with three bugs and one unimplemented function, checked against **18 held-out tests the model never sees**. `sandbox` rows run inside the isolated container used for the ablated model.

MODEL	FIXTURE	HARNESS	VERDICT	MEDIAN	RANGE	HIDDEN TESTS
Sessions still running.						

Every number here is generated from the TSVs in `results/`. Method, caveats and the twelve harness problems found and fixed during this round are in `measurements.md` and `review.md`; exact digests and versions in

results/provenance.txt .